[INCH-POUND]
A-A-55517E
3 June 2025
SUPERSEDING
A-A-55517D
24 June 2020

COMMERCIAL ITEM DESCRIPTION

RESISTOR, FIXED, CARBON FILM,
GENERAL REQUIREMENTS FOR

The General Services Administration has authorized the use of this Commercial Item Description for all federal agencies.

1. SCOPE. This Commercial Item Description (CID) covers the general requirements for a carbon film resistor. These resistor offer long term stability and may be considered as a substitute for carbon composition resistors. Requirements for specific resistors are covered in the individual CID specification sheets. For technical assistance when utilizing parts for parameters not specifically detailed (such as surge or high frequency applications) contact manufacturers listed on applicable CID specification sheets The resistors covered by this CID are intended for commercial/industrial application.

2. CLASSIFICATION OR PART IDENTIFICATION NUMBER (PIN). This CID uses a classification system which is included in the PIN as shown in the following examples (see 7.1).

AA55517	01	***	*
CID number	Applicable CID sheet (two digits)	Resistance value (see 3.3 and 3.4)	Resistance tolerance (see 3.5)

3. SALIENT CHARACTERISTICS.

3.1 Interface and physical dimensions. Resistors supplied to this CID shall be as specified on the applicable CID specification sheet.

3.2 CID specification sheet. The family of carbon, film, resistors shall be in accordance with the requirements specified herein and the applicable CID specification sheet. Unless otherwise noted herein or in the event of a conflict between this general CID and the applicable CID specification sheet, the latter shall govern.

3.3 Resistance. The nominal resistance value expressed in ohms is identified by a three digit number; the first two digits represent significant figures and the last digit specifies the number of zeros to follow. When resistance values less than 10 ohms are required, the letter “R” is substituted for one of the significant digits to represent the decimal point. When the letter “R” is used, significant digits follow.

3.4 Resistance value. Minimum and maximum resistance values shall be as specified (see 3.1).

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data that may improve this document should be sent to: DLA Land and Maritime, ATTN: VAT, Post Office Box 3990, Columbus, OH 43218-3990, or emailed to resistor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A
FSC 5905
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3.5 Resistance tolerance. The resistance tolerance is identified by a single letter in accordance with table I.

TABLE I. Resistance tolerance.

Symbol	Resistance tolerance Percent (±)
G	2
J (Standard)	5
K	10

3.6 Operating ambient temperature. The operating ambient temperature shall be as specified on the applicable CID specification sheet.

3.7 Marking. Resistors shall be legibly and permanently color-coded with the manufacturers standard color coding.

3.8 Recycled, recovered, or environmentally preferable, or biobased materials. Recycled, recovered, or environmentally preferable, or biobased materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.9 Workmanship. Resistor shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Market acceptance. The following market acceptance criteria are necessary to document the quality of the product to be provided under this CID:

a. The company producing the item must have been producing a product meeting the requirements of this CID for at least 2 years.

b. The company producing the item must have sold 1,000 units meeting this CID in the commercial marketplace over the past 2 years.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES

7.1 PIN. The PIN should be used for Government purposes to buy commercial products to this CID. See section 2 for PIN format example.

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7.2 Commercial and Government Entity (CAGE) code. For ordering purposes, inventory control, and submission of these resistors to DLA Land and Maritime under the Military Parts Control Advisory Group (MPCAG) evaluation program, CAGE code 58536 should be used.

7.3 Source of documents.

FEDERAL REGULATIONS

FAR 23.403	-	Federal Acquisition Regulations (FAR) – Use of Recovered Materials

(Copies of these documents are available online at https://www.acquisition.gov/?q=browsefar).

7.4 Ordering data. The contract or order should specify the following:

a. CID document number, revision, and CID PIN.

b. Product conformance provisions.

c. Packaging requirements.

7.5 Government users. To acquire information on obtaining these resistors from the Government inventory system, contact DLA Land and Maritime, ATTN: DLA Land and Maritime-FMX, P.O. Box 3990, Columbus, OH 43218-3990, or telephone (614) 692-3677.

7.6 Changes from previous issue. Marginal notations are used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

MILITARY INTERESTS:

Custodians:
NAVY - EC
DLA - CC

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - 7FXE

Preparing activity:
DLA - CC

Project 5905-2025-038

NOTE: The activities listed above were interested in this document as the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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